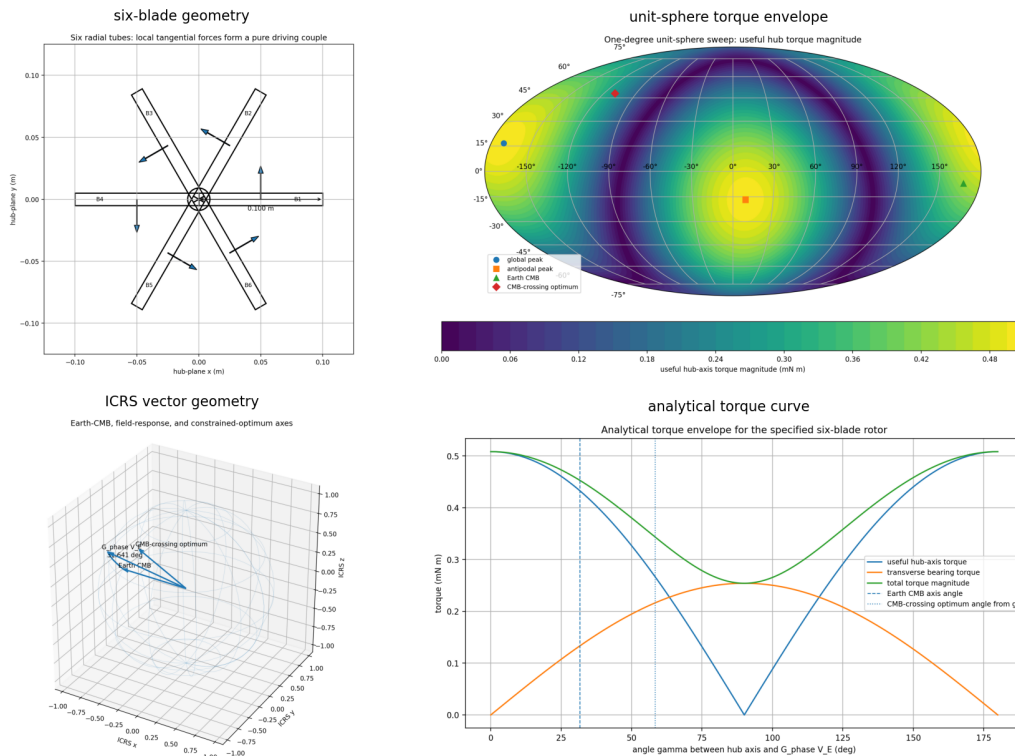


## K-FIELD

# Tubular Rotor Earth-CMB Windmill

Six-blade radial actuator geometry, complete unit-sphere torque envelope, and peak-axis orientation analysis



Title figure. Every panel is generated from the authoritative Cell3 geometry, calibrated Earth-CMB field inputs, exact six-blade torque equations, and the stated tube dimensions and 50,000 rpm spin rate.

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This paper is a standalone educational and archival reference. It defines the K-Field field-response geometry, the rotating-tube lateral-force law, the balanced six-blade windmill architecture, the exact torque vector, the unit-sphere orientation sweep, the global and CMB-constrained peak axes, the mechanical construction ledger, and the machine-reconstruction procedure. All numerical figures and tables are generated from the supplied Cell3 basis, calibrated Earth-CMB inputs, and the stated rotor dimensions.

## Abstract

A long narrow tube spinning about its own longitudinal axis produces a steady lateral K-Field force proportional to its mass and signed axial angular speed. Six such tubes arranged as equally spaced radial blades convert the lateral tube forces into a balanced hub torque. The blade forces sum to zero, while their radial moments add. The useful torque about a hub axis  $h$  is proportional to the scalar projection  $h \cdot (G_{\text{phase}} V_E)$ . Consequently, the complete orientation envelope is a dipolar absolute-cosine surface with two antipodal maxima and a great-circle null.

For six 100 mm long tubes with 10 mm outside diameter, 1 mm wall, nominal stainless density  $8000 \text{ kg/m}^3$ , and axial spin speed 50,000 rpm, the active tube mass is 0.135716803 kg and the first radial mass moment is 0.0067858401 kg m. The exact global peak useful torque is 0.508336676 mN m at hub-axis RA 189.285391 deg, Dec 16.626278 deg and its antipode. The peak is aligned with  $G_{\text{phase}} V_E$ , lies 31.640861 deg from the Earth-CMB vector, and has zero transverse torque.

Requiring every radial blade to pass through the Earth-CMB direction imposes  $h \cdot c = 0$ . The best axis under this constraint is RA 250.207298 deg, Dec 47.867366 deg and its antipode. It produces 0.266669960 mN m of useful torque, 0.216386985 mN m of transverse reaction torque, and 0.343418396 mN m total torque. The constrained useful torque is 52.4593% of the global maximum.

**CENTRAL RESULT:  $|\tau_h| = 0.508336676 |\cos(\gamma)| \text{ mN m}$ , where  $\gamma$  is the angle between the hub axis and  $G_{\text{phase}} V_E$ .**

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## 1. Purpose, scope, and physical question

The device considered here is a radial windmill constructed from six hollow stainless-steel tubes. Each tube is both a structural blade and an independently rotating mass. Its long axis lies radially in the hub plane, while the tube spins about that same long axis. The common hub is free to rotate about an axis normal to the blade plane.

Reversing a tube spin reverses its lateral K-Field force by exactly 180 deg. The windmill architecture uses that parity to arrange opposite blades so that their force vectors are opposite, canceling translation, while their moments about the hub have the same sign and add. The analysis therefore addresses four linked questions: the exact torque produced by the specified hardware; the dependence of that torque on hub-axis orientation; the relationship of the peak axes to the Earth-CMB traversal vector; and the distinction between the unconstrained maximum and the best orientation that permits actual CMB crossings by the radial blades.

The reported envelope is the zero-to-moderate-hub-speed drive envelope. It applies the calibrated tube-force law instantaneously to each blade and neglects compound terms that depend on hub speed. Those terms enter only after the hub has acquired appreciable angular velocity and are treated as the next-order torque-speed problem.

### 1.1 Coordinate and sign conventions

All vectors are expressed in the ICRS/J2000 Cartesian frame. Right ascension and declination are used only as labels for unit-vector directions. The Earth-CMB traversal unit vector is denoted  $\hat{c}$ . The anisotropically transformed Earth velocity is  $\mathbf{g} = G_{\text{phase}} \mathbf{V}_E$ . The hub-axis unit vector is  $\hat{h}$ . A blade-axis unit vector  $\hat{n}$  lies in the plane perpendicular to  $\hat{h}$ , and the positive local hub tangent is  $\hat{t} = \hat{h} \times \hat{n}$ .

Each tube is assigned the same outward-viewed spin handedness. For an opposite blade, outward direction reverses; therefore its global spin vector also reverses. This is the required parity for equal-and-opposite blade forces and equal hub torques. Reversing all six tube spins reverses the signed torque everywhere without changing the magnitude envelope.

## 2. Cell3 geometric and field-response foundation

The force model is anchored to the supplied Cell3 geometry and its physical response matrix. Cell3 is defined by three unequal non-orthogonal basis vectors  $\mathbf{a}$ ,  $\mathbf{b}$ , and  $\mathbf{c}$ . Their Gram geometry supplies three edge lengths, three included angles, three primitive face areas, and one primitive volume. The same basis generates the 27 centered address nodes used throughout the K-Field construction.

Dimensionally correct centered Cell3 and all 27 address nodes

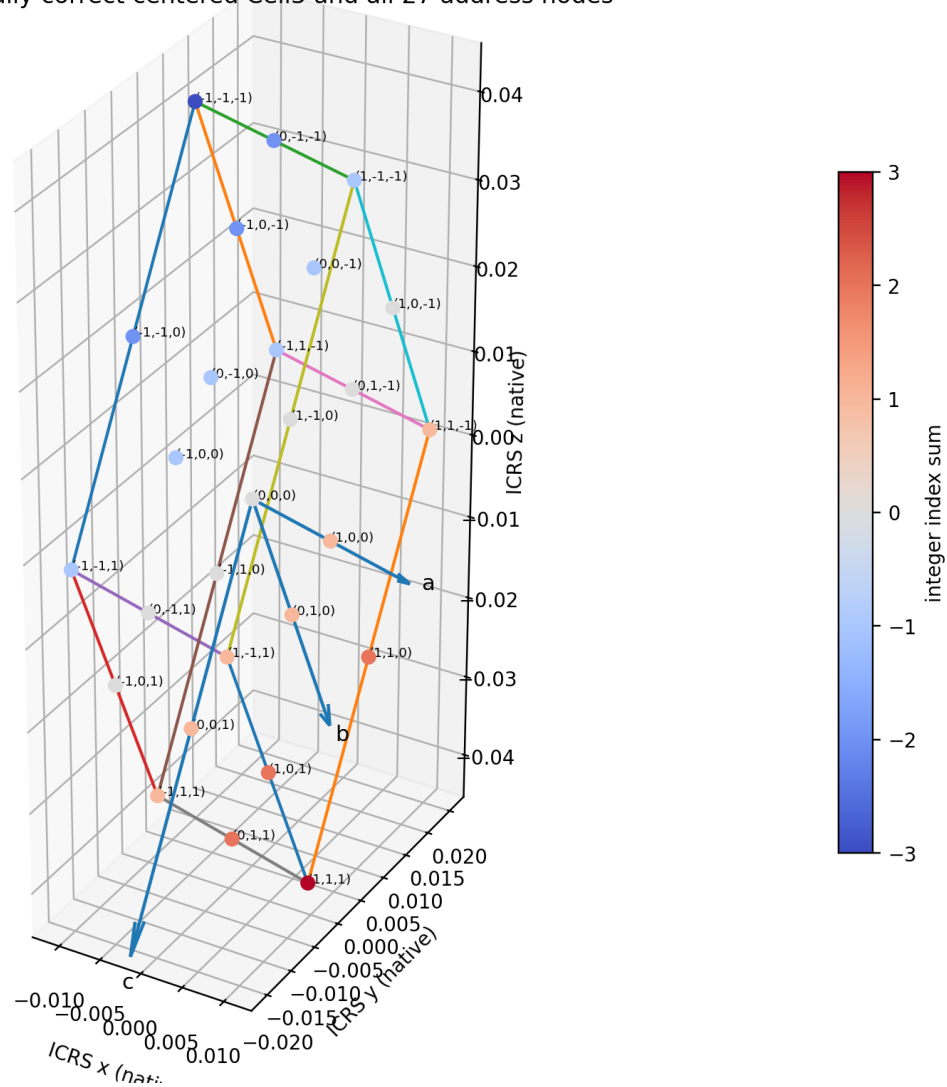


Figure 1. Dimensionally correct centered Cell3 in native ICRS coordinates. Every node has integer address  $(i, j, k)$  with  $i, j, k$  in  $\{-1, 0, 1\}$ ; its coordinate is  $(i a + j b + k c)/2$ . The eight corner nodes and twelve connecting edges define the centered base cell.

## 2.1 Basis and primitive measures

| Quantity         | Value             | Dimension           |
|------------------|-------------------|---------------------|
| $ a $            | 0.0207970587824   | native              |
| $ b $            | 0.0364942051307   | native              |
| $ c $            | 0.0493085658999   | native              |
| $A_{ab}$         | 0.000754721789673 | native <sup>2</sup> |
| $A_{ac}$         | 0.00101125575769  | native <sup>2</sup> |
| $A_{bc}$         | 0.00138578443173  | native <sup>2</sup> |
| primitive volume | 2.84206693781e-05 | native <sup>3</sup> |

## 2.2 Primitive face planes

Each face family consists of two parallel planes. Their unit normals are generated directly from the corresponding basis-vector cross products. The centered-cell boundary equations have the form  $\hat{n} \cdot \mathbf{r} = d_{\text{plus}}$  and  $\hat{n} \cdot \mathbf{r} = d_{\text{minus}}$ . The six planes and the 27 nodes establish the geometric orientation inherited by  $G_{\text{phase}}$ .

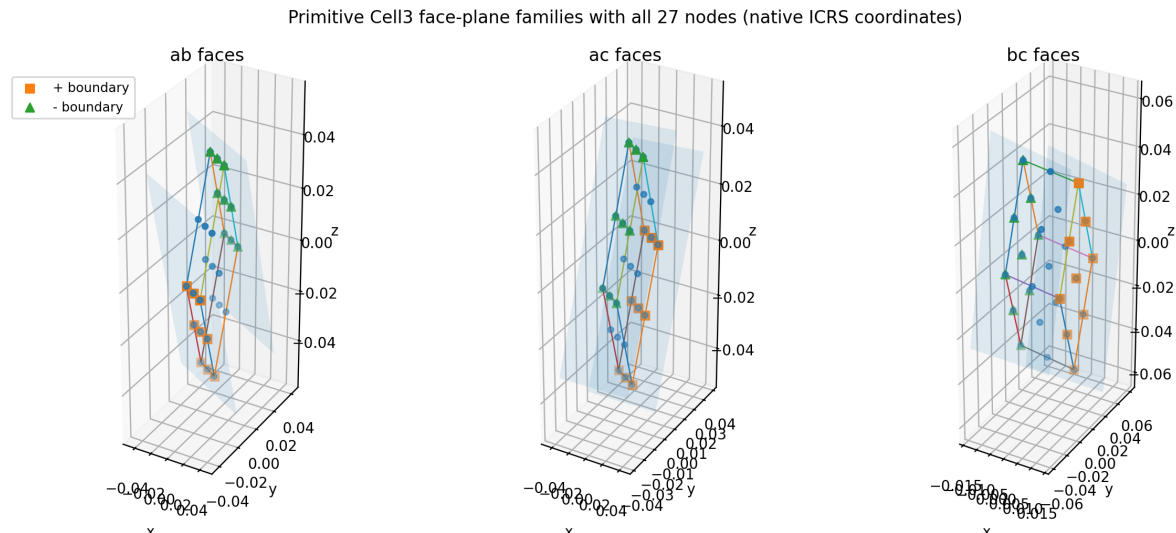


Figure 2. The ab, ac, and bc face-plane families. Square and triangular markers identify the nodes on the two boundary planes of each family. Plane locations and node coordinates are expressed in the same native ICRS frame.

## 2.3 Earth-CMB and response matrix

| Input                            | x                  | y                  | z                 | Norm / note                          |
|----------------------------------|--------------------|--------------------|-------------------|--------------------------------------|
| Earth velocity $V_E$ (m/s)       | -3.59003199066e+05 | 76679.886034       | -44701.0601007    | 3.69812502000e+05                    |
| $g = G_{\text{phase}} V_E$ (m/s) | -8.80221981635e+05 | -1.43911436562e+05 | 2.66335026949e+05 | 9.30825217270e+05                    |
| Earth-CMB unit vector            | -0.970770855836    | 0.207348009111     | -0.120874929482   | RA 167.943300 deg; Dec -6.942600 deg |

The response matrix is symmetric and anisotropic:

| row/col | 1             | 2             | 3             |
|---------|---------------|---------------|---------------|
| 1       | 2.9764106378  | 1.2431507795  | -2.0803600354 |
| 2       | 1.2431507795  | 2.5922286635  | -2.3178839937 |
| 3       | -2.0803600354 | -2.3178839937 | 6.7735709133  |

$$g = G_{\text{phase}} V_E$$

The direction of  $g$  is RA 189.285391 deg, Dec 16.626278 deg. Its angular separation from the positive Earth-CMB traversal vector is 31.640861 deg. This separation controls the difference between the global torque optimum and the CMB-crossing constrained optimum.

## Earth-CMB, field-response, and constrained-optimum axes

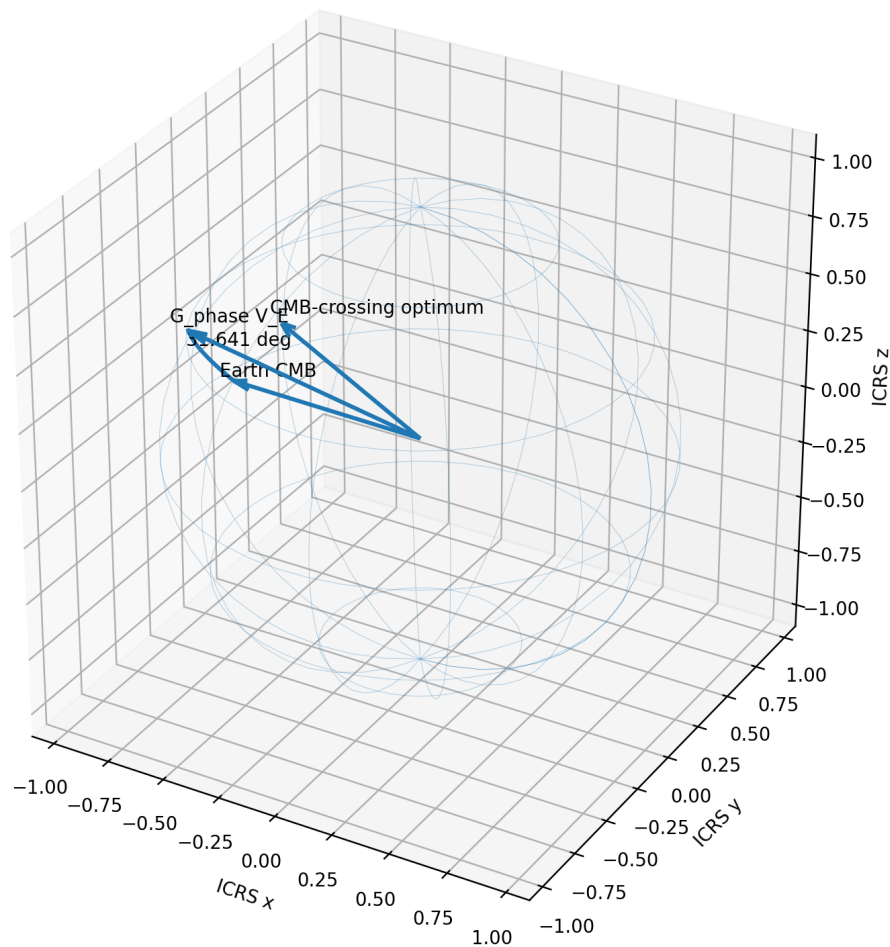


Figure 3. Unit-sphere vector geometry. The global peak hub axis follows  $G\_phase V\_E$ , while the best CMB-crossing axis is the projection of  $G\_phase V\_E$  into the plane perpendicular to the Earth-CMB vector.

### 3. Six-blade tubular windmill construction

The active rotor consists of six identical hollow tubes equally spaced by 60 deg. Each tube extends radially from the hub axis to a centerline length of 0.100 m. The outside diameter is 0.010 m, the wall thickness is 0.001 m, and the resulting inside diameter is 0.008 m. The nominal material density is 8000 kg/m<sup>3</sup>. No hub-root offset is included in the reference values; a finite root radius is treated separately.



Six-blade construction dimensions and material section

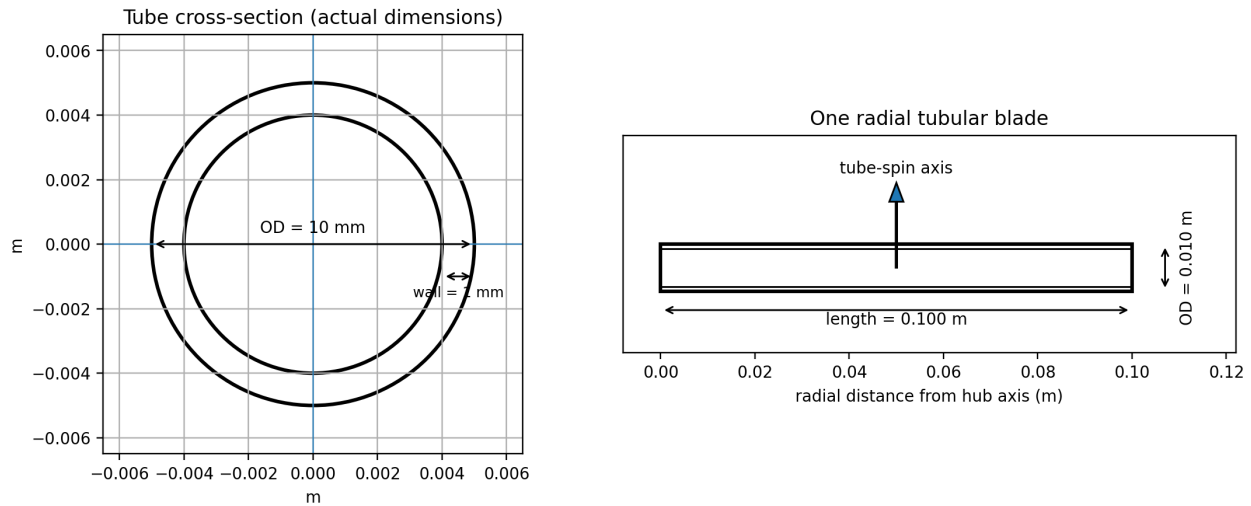


Figure 4. Dimensioned tube cross-section and side view. The plotted dimensions are the actual dimensions used in the mass, inertia, spin-energy, and torque calculations.

### 3.1 Mass ledger

$$A_{\text{steel}} = \pi (R_o^2 - R_i^2)$$

$$m_{\text{blade}} = \rho A_{\text{steel}} L$$

$$M = 6 m_{\text{blade}}$$

| Quantity                         | Exact calculated value | Unit           |
|----------------------------------|------------------------|----------------|
| Outside radius $R_o$             | 0.005                  | m              |
| Inside radius $R_i$              | 0.004                  | m              |
| Wall thickness                   | 0.001                  | m              |
| Steel area                       | 2.8274338823e-05       | m <sup>2</sup> |
| Volume per blade                 | 2.8274338823e-06       | m <sup>3</sup> |
| Mass per blade                   | 0.0226194671058        | kg             |
| Total active tube mass           | 0.135716802635         | kg             |
| Linear mass density              | 0.226194671058         | kg/m           |
| Mean radial lever arm            | 0.05                   | m              |
| First radial mass moment $\mu_1$ | 0.00678584013175       | kg m           |

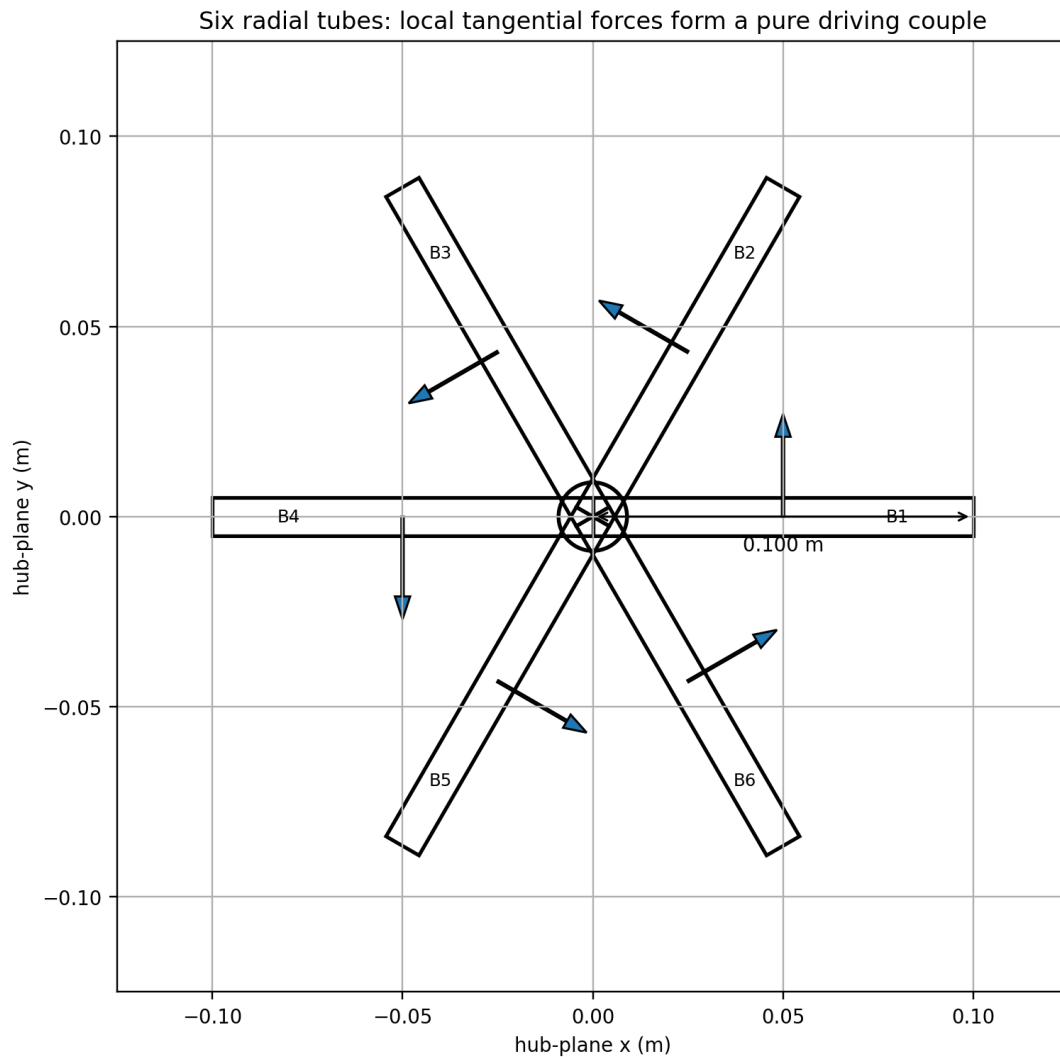


Figure 5. Six-blade plan view in the hub plane. Each arrow is the local tangential force direction for a common outward-viewed tube-spin handedness. The vector sum of the six blade forces is zero, while the radial moments add.

## 4. Tube-spin kinematics at 50,000 rpm

The tube spin rate is 50,000 revolutions per minute, equal to 833.333333 revolutions per second. The signed angular speed is 5235.987756 rad/s. The outer cylindrical surface travels at 26.179939 m/s and the inner surface at 20.943951 m/s.

| omega_s = RPM ( 2 pi / 60 )    |                   |       |
|--------------------------------|-------------------|-------|
| Quantity                       | Value             | Unit  |
| Spin speed                     | 50000             | rpm   |
| Spin frequency                 | 833.33333333      | Hz    |
| Angular speed omega_s          | 5235.98775598     | rad/s |
| Spin period                    | 0.0012            | s     |
| Outer surface speed            | 26.1799387799     | m/s   |
| Inner surface speed            | 20.9439510239     | m/s   |
| Outer centripetal acceleration | 1.37077838904e+05 | m/s^2 |

| Quantity                       | Value             | Unit             |
|--------------------------------|-------------------|------------------|
| Inner centripetal acceleration | 1.09662271123e+05 | m/s <sup>2</sup> |

## 4.1 Axial rotational energy and angular momentum

$$I_{\text{spin,blade}} = (m_{\text{blade}}/2) (R_o^2 + R_i^2)$$

$$E_{\text{spin}} = (1/2) I_{\text{spin}} \omega_s^2 ; H_{\text{spin}} = I_{\text{spin}} \omega_s$$

| Quantity                                   | Per blade         | Six-blade scalar sum | Unit                 |
|--------------------------------------------|-------------------|----------------------|----------------------|
| Axial moment of inertia                    | 4.63699075670e-07 | 2.78219445402e-06    | kg m <sup>2</sup>    |
| Rotational energy                          | 6.35628671946     | 38.1377203168        | J                    |
| Axial angular momentum magnitude           | 0.00242792268267  | 0.014567536096       | kg m <sup>2</sup> /s |
| Ideal rotational stress scale $\rho v_o^2$ | 5.48311355616e+06 | same for material    | Pa                   |

The vector sum of tube spin angular momenta is zero in the perfectly symmetric six-blade state, because the six radial spin axes sum to zero. The scalar sum remains relevant to local bearing loads and stored energy.

## 5. Integrated lateral-force law for one tube

A circular tube cross-section contains every circumferential phase simultaneously. Opposite mass elements have opposite radial and tangential directions; both signs reverse in the force kernel, so the paired forces add rather than cancel. Integrating around the cross-section removes the rotating force pulse and leaves a steady lateral force. Axial integration contributes only total tube mass.

$$F_{\text{tube}} = (\kappa_F m \omega_s / 2) \hat{n} \text{ cross } (G_{\text{phase}} V_E)$$

The force is linear in signed tube angular speed. Reversing  $\omega_s$  changes  $F_{\text{tube}}$  to  $-F_{\text{tube}}$  with unchanged magnitude. At fixed  $\omega_s$ , the force is independent of tube radius and length except through total mass. Tube radius controls stored rotational energy, stress, and achievable angular speed.

### 5.1 Tangential component relative to the hub

For a radial blade  $\hat{n}$  and hub tangent  $\hat{t} = \hat{h} \text{ cross } \hat{n}$ , the tangential force component is independent of blade azimuth:

$$F_t = F_{\text{tube}} \cdot \hat{t} = -(\kappa_F m \omega_s / 2) (\hat{h} \cdot \hat{g})$$

Every blade therefore receives the same signed tangential force relative to its own local tangent. The torque is continuous around the full hub revolution; it is not confined to the two instants at which a blade crosses the Earth-CMB vector.

## 6. Opposite blades and the pure-force-couple condition

For a diametrically opposite blade,  $\hat{n}_2 = -\hat{n}_1$  and  $\hat{r}_2 = -\hat{r}_1$ . The same outward-viewed spin handedness gives opposite global spin vectors and therefore opposite K-Field forces:  $F_2 = -F_1$ . Their translational sum is zero. Their torques are equal because  $(-\hat{r}_1) \text{ cross } (-F_1) = \hat{r}_1 \text{ cross } F_1$ .

Diametrically opposite blades: zero net force and additive torque

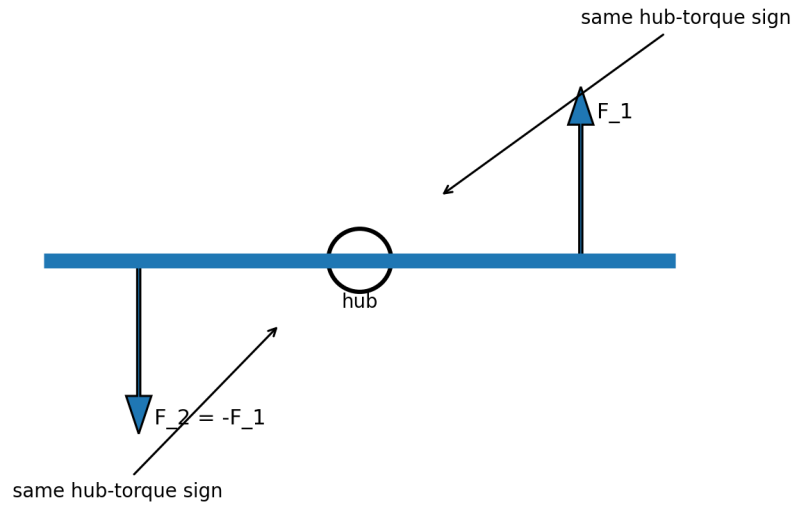


Figure 6. Opposite tube blades produce equal-and-opposite forces and equal hub torques. The pair is a pure couple: net translation cancels while angular drive remains.

**For the balanced six-blade array:  $F_{\text{net}} = 0$  while  $\tau_h$  is nonzero.**

A continuous through-shaft that forces opposite blades to share the same global angular-velocity vector would impose the wrong parity and cancel the desired torque. Each radial tube must preserve the common outward-viewed handedness, either through independent drives or a transmission that reverses global spin across the hub.

## 7. Exact torque of a symmetric six-blade array

A mass element  $dm$  at radius  $s$  on blade  $n_{\text{hat}}$  has position  $r = s n_{\text{hat}}$ . Its lateral force is  $dF = (\kappa_F \omega s/2) n_{\text{hat}} \text{ cross } g \, dm$ . The elemental torque is

$$d\tau = (\kappa_F \omega s/2) s [n_{\text{hat}} (n_{\text{hat}} \cdot g) - g] \, dm$$

Define the complete first radial mass moment  $\mu_1 = \sum \int s \, dm$ . For six equally spaced equal blades, the second directional moment is

$$\sum n_i n_i^T = (N/2) (I - h_{\text{hat}} h_{\text{hat}}^T)$$

Substitution gives the complete torque vector:

$$\tau(h_{\text{hat}}) = -(\kappa_F \omega s \mu_1 / 4) [g + (h_{\text{hat}} \cdot g) h_{\text{hat}}]$$

Projection onto the constrained hub axis gives the useful drive torque:

$$\tau_h = h_{\text{hat}} \cdot \tau = -(\kappa_F \omega s \mu_1 / 2) (h_{\text{hat}} \cdot g)$$

The component perpendicular to the hub axis is

$$\tau_{\text{perp}} = -(\kappa_F \omega_s \mu_1 / 4) [g - (h_{\text{hat}} \cdot g) h_{\text{hat}}]$$

The transverse term is a bearing-reaction torque. It vanishes only when  $h_{\text{hat}}$  is parallel or antiparallel to  $g$ . A fixed-axis bearing structure must carry this component; a freely gimbaled assembly experiences an orientation-changing moment.

## 8. Analytical torque envelope on the unit sphere

Let  $\gamma$  be the angle between the hub axis and  $g$ . Since  $h_{\text{hat}} \cdot g = |g| \cos \gamma$ , the envelope reduces to three scalar functions.

$$|\tau_h| = \tau_{\text{max}} |\cos \gamma| ; \tau_{\text{max}} = 5.083366763745\text{e-}04 \text{ N m}$$

$$|\tau_{\text{perp}}| = (\tau_{\text{max}}/2) |\sin \gamma|$$

$$|\tau| = (\tau_{\text{max}}/2) \sqrt{1 + 3 \cos^2 \gamma}$$

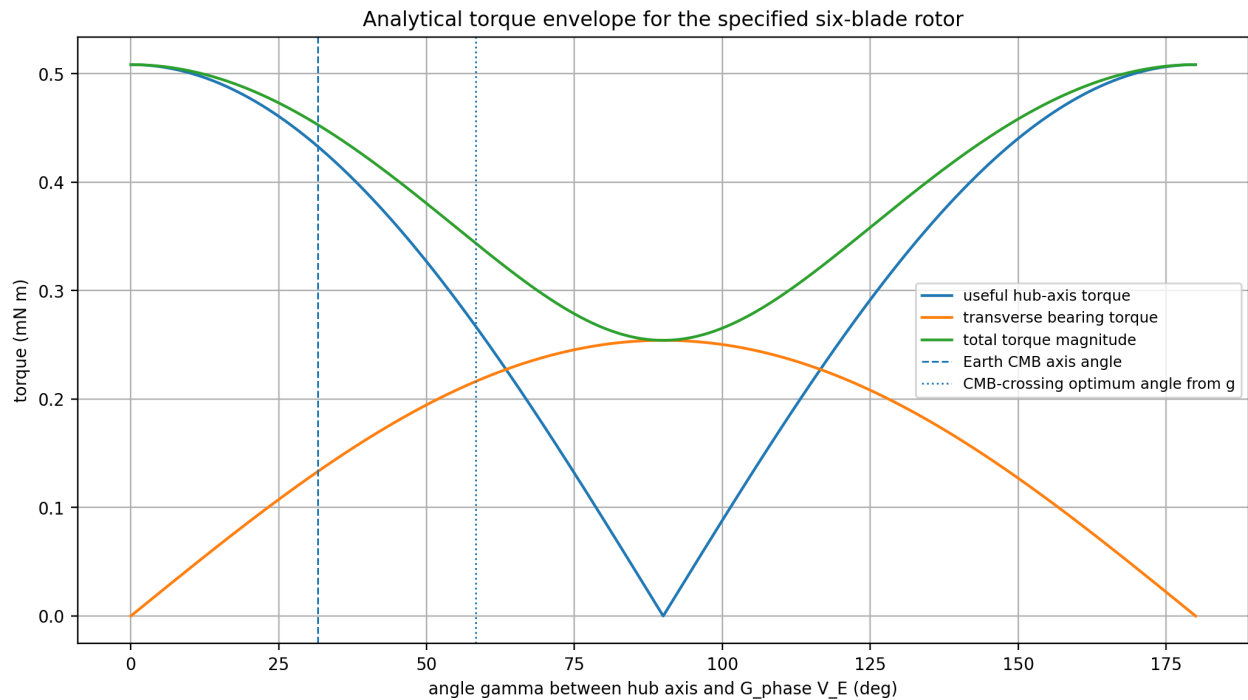


Figure 7. Analytical torque envelope. Useful hub torque follows an absolute cosine, transverse torque follows one-half sine, and total torque follows the vector combination. The global maximum occurs at  $\gamma = 0$  or  $180$  deg; the useful-torque null occurs at  $\gamma = 90$  deg.

The exact maximum useful torque is  $5.083366763745\text{e-}04 \text{ N m} = 0.508336676 \text{ mN m}$ . The Earth-calibration interval gives  $0.503158979$  to  $0.513514374 \text{ mN m}$ . At  $\gamma = 90$  deg, useful torque is zero but transverse torque remains  $0.254168338 \text{ mN m}$ .

## 9. One-degree hub-axis sweep

The numerical sweep samples right ascension from  $0$  through  $359$  deg in  $1$  deg increments and declination from  $-90$  through  $+90$  deg in  $1$  deg increments, producing  $65,160$  directed-axis samples. At every point, the calculation evaluates  $h_{\text{hat}}$ ,  $h_{\text{hat}} \cdot g$ , the signed and absolute hub torque, transverse torque, total torque, angular separation from  $g$ , and angular separation from the Earth-CMB vector.

The RA-Dec grid is a coordinate sweep rather than an equal-area statistical quadrature; points near the poles are more densely represented per unit solid angle. This does not alter the location or value of extrema because the analytical

envelope is evaluated pointwise. The machine-readable companion file preserves the complete grid in columnar arrays.

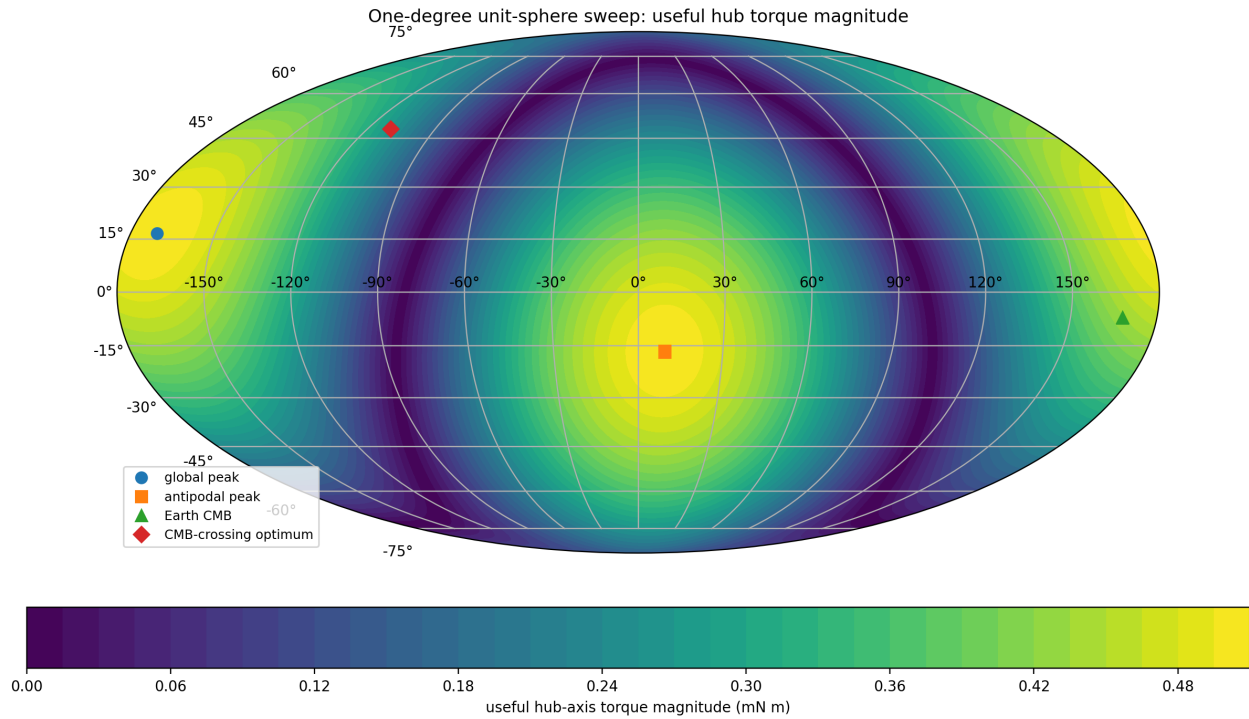


Figure 8. Numerical one-degree sweep of useful torque magnitude. The two antipodal yellow maxima follow the  $G_{\text{phase}} V_E$  axis line. The dark great-circle belt is the useful-torque null.

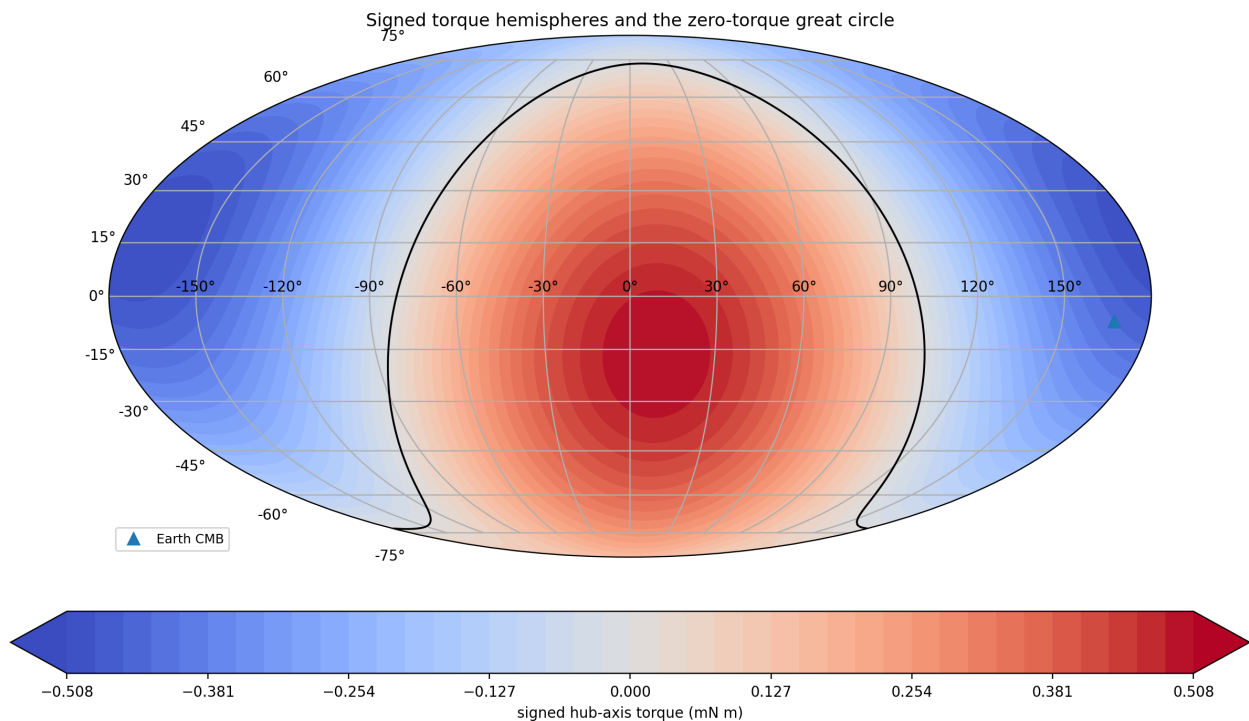


Figure 9. Signed hub torque. The zero contour is a great circle perpendicular to  $G_{\text{phase}} V_E$ . The two hemispheres drive opposite signed rotation for a fixed tube-spin convention.

## 9.1 Numerical extrema and agreement with the closed form

| Result                     | RA (deg)      | Dec (deg)     | Useful torque (mN m) | Relative to exact peak |
|----------------------------|---------------|---------------|----------------------|------------------------|
| Exact positive peak axis   | 189.285391176 | 16.626277960  | 0.508336676375       | 1.000000               |
| Exact antipodal axis       | 9.285391176   | -16.626277960 | 0.508336676375       | 1.000000               |
| Nearest 1-deg grid maximum | 9             | -17           | 0.508320084366       | 0.99996736             |
| Nearest 1-deg grid null    | 324           | 67            | 0.000016967972       | grid-dependent         |

## 10. Global peak torque axis orientations

Maximum useful torque occurs when the hub axis is parallel or antiparallel to  $\mathbf{g} = G_{\text{phase}} \mathbf{V}_E$ . The physical axis is one unoriented line; the two RA-Dec directions are antipodal coordinate representations. The complete torque vector is parallel to the hub axis, so the transverse bearing torque is exactly zero.

| Axis representation | Unit vector x   | Unit vector y  | Unit vector z  | RA (deg)      | Dec (deg)     |
|---------------------|-----------------|----------------|----------------|---------------|---------------|
| +g_hat              | -0.945636157362 | -0.15460629331 | 0.28612785946  | 189.285391176 | 16.626277960  |
| -g_hat              | 0.945636157362  | 0.15460629331  | -0.28612785946 | 9.285391176   | -16.626277960 |

At the peak, each blade produces 1.694455588 mN of tangential force. Six such forces form a balanced circular couple. Their net translational force is zero, and their total useful torque is 0.508336676 mN m.

Reversing all tube spins changes the torque vector by 180 deg. The axis locations and torque magnitudes are unchanged. Reversing only selected blades destroys force balance and introduces translational and cyclic bearing loads.

## 11. Relationship to the Earth-CMB traversal vector

The Earth-CMB traversal direction is RA 167.943299880 deg, Dec -6.942599922 deg. The global peak hub axis is separated from this vector by 31.640861271 deg. Therefore, aligning the hub axis with the maximum-torque direction does not place the CMB vector in the blade plane.

At the global maximum, every blade axis is confined to the plane perpendicular to  $\mathbf{g\_hat}$ . The smallest angle any blade can make with the Earth-CMB vector is 58.359138729 deg. Actual blade-CMB alignment is therefore impossible in the global-maximum orientation.

Aligning the hub axis itself with the CMB vector produces 0.432773970 mN m of useful torque, which is 85.1353% of the global maximum. However, the blade plane is then perpendicular to CMB, so no radial tube crosses the CMB direction.

## 12. Peak torque subject to actual CMB crossing

For every blade to pass through the CMB direction once per half revolution, the Earth-CMB vector must lie in the hub plane. The constraint is

$$\mathbf{h\_hat} \cdot \mathbf{c\_hat} = 0$$

Among all hub axes satisfying this constraint, useful torque is maximized by projecting  $\mathbf{g}$  into the plane perpendicular to  $\mathbf{c\_hat}$ :

$$\mathbf{h\_cross} = [\mathbf{g} - (\mathbf{c\_hat} \cdot \mathbf{g})\mathbf{c\_hat}] / |\mathbf{g} - (\mathbf{c\_hat} \cdot \mathbf{g})\mathbf{c\_hat}|$$

## CMB-crossing constraint: CMB lies in the hub plane

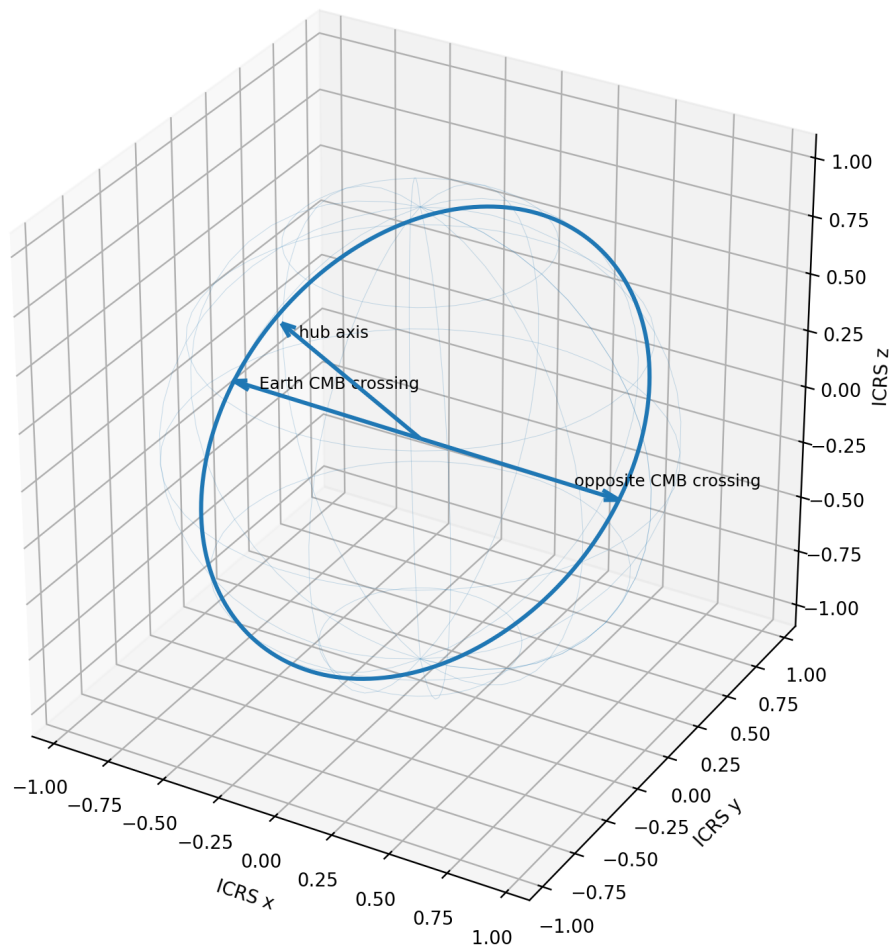


Figure 10. The constrained-optimum hub plane is the great circle perpendicular to  $h_{\text{cross}}$ . It contains both the positive and negative Earth-CMB directions, so each radial blade crosses the CMB line twice per hub revolution.

| Axis representation | RA (deg)      | Dec (deg)     | Useful torque (mN m) | Transverse torque (mN m) | Total torque (mN m) |
|---------------------|---------------|---------------|----------------------|--------------------------|---------------------|
| $h_{\text{cross}}$  | 250.207297827 | 47.867366475  | 0.266669960389       | 0.216386984806           | 0.343418396373      |
| $-h_{\text{cross}}$ | 70.207297827  | -47.867366475 | 0.266669960389       | 0.216386984806           | 0.343418396373      |

The CMB-crossing useful torque is 52.4593% of the global maximum. Its calibration interval is 0.263953775 to 0.269386145 mN m. Each blade contributes 0.888899868 mN of constant tangential force.

The transverse moment is not a torque ripple; it is a steady vector component determined by the hub-axis orientation. A rigid bearing frame carries it as a reaction moment. A gimballed frame responds by reorienting or precessing toward the zero-transverse global optimum.



### 13. Complete reference-orientation comparison

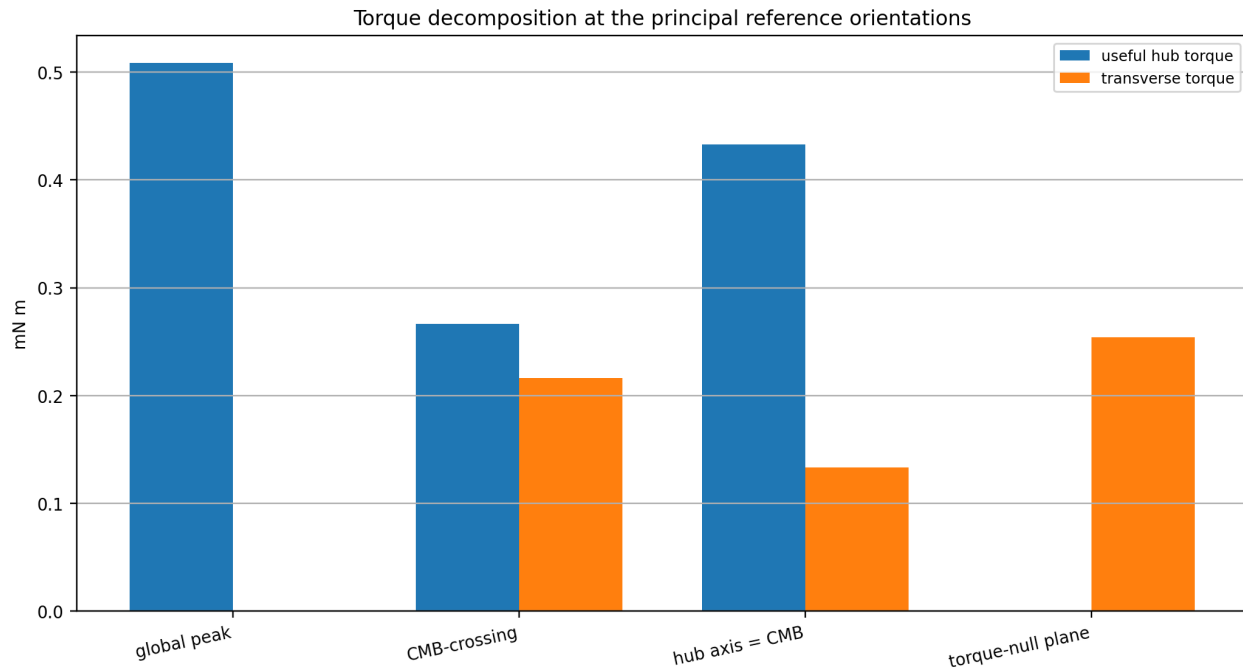


Figure 11. Useful and transverse torque components at the four principal orientation classes. The global peak is the only nonzero useful-torque orientation with no transverse component.

| Hub-axis condition                                           | Axis RA / Dec              | Useful torque    | Transverse torque | Total torque     | CMB crossing?         |
|--------------------------------------------------------------|----------------------------|------------------|-------------------|------------------|-----------------------|
| Hub axis parallel to G_phase V_E                             | 189.285391 / 16.626278 deg | 0.508336676 mN m | 0.000000000 mN m  | 0.508336676 mN m | False                 |
| Hub axis perpendicular to CMB and optimized for CMB crossing | 250.207298 / 47.867366 deg | 0.266669960 mN m | 0.216386985 mN m  | 0.343418396 mN m | True                  |
| Hub axis parallel to Earth CMB vector                        | 167.943300 / -6.942600 deg | 0.432773970 mN m | 0.133334980 mN m  | 0.452848237 mN m | False                 |
| Hub axis perpendicular to G_phase V_E                        | great-circle family        | 0.000000000 mN m | 0.254168338 mN m  | 0.254168338 mN m | orientation-dependent |

The hub-axis line is antipodally redundant. Replacing  $\hat{h}$  by  $-\hat{h}$  leaves the physical torque vector invariant but reverses the sign of the scalar torque when measured against the newly reversed positive-axis convention. This is a coordinate-sign change, not a second mechanical configuration.

### 14. Initial hub dynamics and angular acceleration

The hub-axis moment of inertia includes the radial length of each tube and the finite circular cross-section. For one thick-walled tube whose inner end lies on the hub axis:

$$I_{h,blade} = m [L^2/3 + (R_o^2 + R_i^2)/4]$$

The six-blade active-tube inertia is  $4.537804393439e-04 \text{ kg m}^2$ . Neglecting hub, motor, transmission, bearing, and payload inertia, the initial angular accelerations are  $1.120226066 \text{ rad/s}^2$  at the global optimum and  $0.587662969 \text{ rad/s}^2$  at the CMB-crossing optimum.

| Orientation    | Torque (N m)      | Active-tube $I_h$ (kg m <sup>2</sup> ) | Initial alpha (rad/s <sup>2</sup> ) |
|----------------|-------------------|----------------------------------------|-------------------------------------|
| Global maximum | 0.000508336676375 | 0.000453780439344                      | 1.12022606596                       |

| Orientation          | Torque (N m)      | Active-tube $I_h$ (kg m <sup>2</sup> ) | Initial alpha (rad/s <sup>2</sup> ) |
|----------------------|-------------------|----------------------------------------|-------------------------------------|
| CMB-crossing maximum | 0.000266669960389 | 0.000453780439344                      | 0.587662969288                      |

Actual angular acceleration is obtained by replacing  $I_h$  with the complete assembled inertia. The drive torque remains determined by the active-tube first radial mass moment, while inactive hub and motor mass increases inertia without increasing K-Field torque. Maximizing active-mass fraction is therefore a primary design objective.

## 15. Finite hub radius and lever-arm scaling

The reference calculation places the inner tube end on the hub axis. If the tube begins at a finite root radius  $r_0$  while its length and mass remain unchanged, its mean lever arm becomes  $r_0 + L/2$ . All torque values scale by

$$\text{torque}(r_0) / \text{torque}(0) = (r_0 + L/2) / (L/2)$$

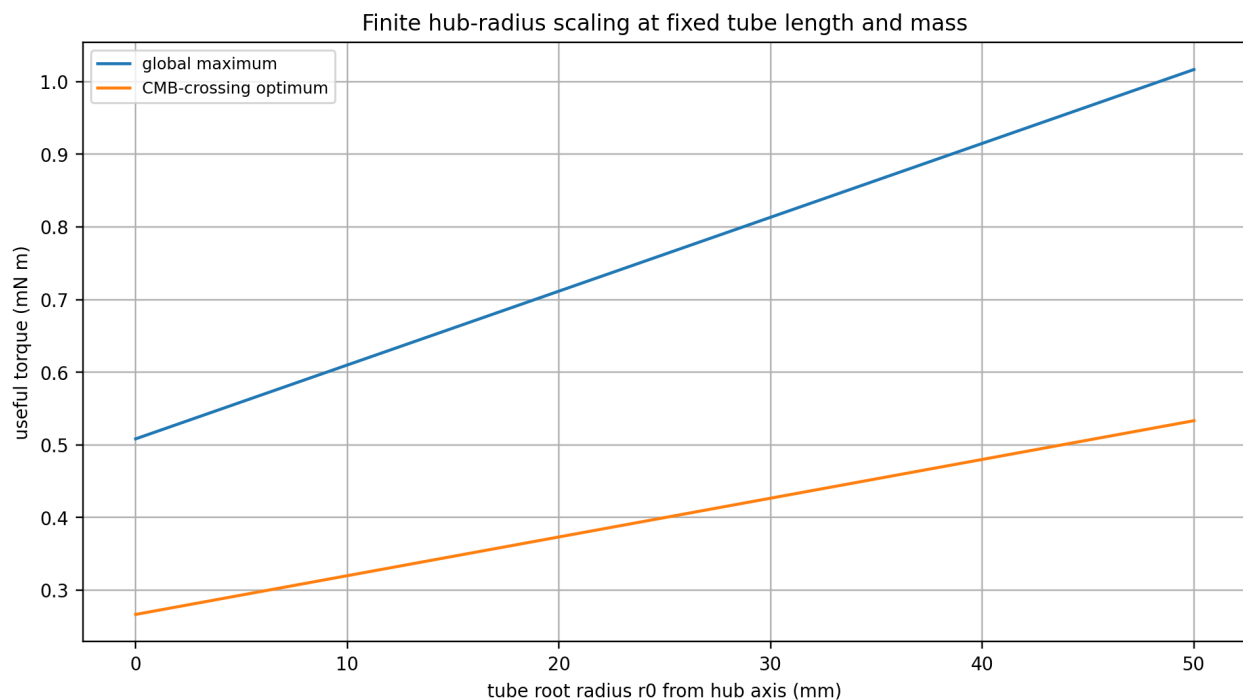


Figure 12. Useful torque rises linearly with tube root radius because the first radial mass moment increases. The plotted mass and tube length remain fixed.

The gain in torque must be compared with the increase in hub-axis inertia. A larger root radius generally increases torque linearly but increases inertia through the square of radius. It therefore favors stall torque more strongly than angular acceleration.

## 16. Mechanical load interpretation at 50,000 rpm

The tube axial spin produces an outer-surface speed of 26.18 m/s and an ideal rotational stress scale of approximately 5.48 MPa. This scale is modest relative to common stainless-steel strength levels, but practical qualification is controlled by balance, straightness, bearing speed, joint design, local stress concentration, material certification, and fatigue rather than by the ideal uniform-ring stress alone.

Each tube also carries gyroscopic bearing loads when the hub rotates. Its axial spin angular momentum is radial, so hub rotation continually changes its direction. The local gyroscopic moment magnitude is approximately  $I_{\text{spin,blade}} \omega_a \Omega_h$ . The six vector moments cancel globally in the perfectly symmetric state, but each blade support must carry its

local load.

$$\tau_{\text{gyro,blade}} \text{ approximately } I_{\text{spin,blade}} \omega_s \Omega_h$$

The narrow tube radius is advantageous twice: it permits high angular speed at moderate surface speed, and it reduces axial spin inertia and gyroscopic load in proportion to radius squared at fixed mass and angular speed.

## 17. Torque, power, and direction reversal

At hub angular speed  $\Omega_h$ , mechanical power delivered about the constrained axis is

$$P_h = \tau_h \Omega_h$$

When the hub rotates in the driven direction, the torque performs positive work. When externally driven in the opposite direction, the same torque performs negative work. Reversing all six tube spins changes  $\tau_h$  to  $-\tau_h$  and reverses the preferred hub direction.

Because the six translational forces cancel, the device does not require a net linear reaction to generate the ideal hub torque. Bearing reactions arise from the transverse torque component, manufacturing asymmetry, gyroscopic loads, and compound-rotation corrections.

## 18. Compound-rotation extension and torque-speed regime

The present unit-sphere envelope is the drive-torque boundary at zero or moderate hub speed. A complete rotating machine adds hub velocity to every tube mass element:

$$\mathbf{v} = \mathbf{V}_E + \Omega_h \text{ cross } \mathbf{r} + \omega_s \text{ cross } \boldsymbol{\rho}$$

Substitution into the full force kernel produces terms proportional to  $\Omega_h$ ,  $\omega_s \Omega_h$ , and  $\Omega_h^2$  in addition to the established linear  $\omega_s$  drive term. Symmetry removes many common-mode contributions, but the remaining terms determine the torque-speed curve, backreaction, saturation, and steady operating speed.

The exact zero-speed envelope derived here remains the correct orientation map and startup torque. The next machine-level calculation is a full double integration over blade length and cross-section while simultaneously advancing hub and tube phases.

## 19. Experimental realization and measurement protocol

A rigorous experiment uses a stiff six-arm hub with six matched tube rotors, independent or parity-corrected drives, and a low-friction hub bearing. Tube masses, polar inertias, spin rates, and outward-viewed handedness must be matched. The assembly is oriented by a two-axis gimbal whose axes are registered to ICRS through local time, geodetic location, and an astronomical attitude solution.

The principal measurement is signed hub torque as a function of orientation. A torsion balance, calibrated torque transducer, or measured angular acceleration can be used. The predicted orientation signature is more diagnostic than one static reading: two antipodal maxima on the  $G_{\text{phase}} V_E$  line, a zero-torque great circle, sign reversal across that great circle, and linear reversal when all tube spins reverse.

## 19.1 Minimum orientation sequence

| Step | Hub-axis condition                            | Expected useful torque                        |
|------|-----------------------------------------------|-----------------------------------------------|
| 1    | Global positive peak axis                     | 0.508336676 mN m                              |
| 2    | Antipodal peak-axis representation            | same magnitude; axis sign convention reversed |
| 3    | Any point on $h \cdot g = 0$ great circle     | zero useful torque                            |
| 4    | Earth-CMB axis                                | 0.432773970 mN m                              |
| 5    | CMB-crossing optimum                          | 0.266669960 mN m                              |
| 6    | Repeat steps 1-5 with all tube spins reversed | all signed useful torques reverse             |

A complete sphere sweep is not required for the first test, but a dense orientation map strongly separates the predicted dipolar envelope from bearing bias, gravitational imbalance, motor torque ripple, and laboratory-frame thermal drift.

## 20. Principal findings

1. The six specified tubes contain 0.135716803 kg of active stainless-steel mass and have first radial mass moment 0.0067858401 kg m.
2. At 50,000 rpm, the global peak useful torque is 0.508336676 mN m.
3. The peak hub-axis line is RA 189.285391 deg, Dec 16.626278 deg and its antipode, exactly parallel to G\_phase V\_E.
4. The useful-torque envelope is an absolute cosine of angular separation from G\_phase V\_E; the null is the perpendicular great circle.
5. The global peak lies 31.640861 deg from the Earth-CMB vector and does not permit blade-CMB alignment.
6. The best CMB-crossing axis is RA 250.207298 deg, Dec 47.867366 deg and its antipode.
7. The CMB-crossing orientation produces 0.266669960 mN m useful torque and 0.216386985 mN m transverse torque.
8. Sixfold symmetry cancels net translation and tube spin angular momentum while retaining a steady hub torque.
9. Reversing every tube spin reverses the torque direction by 180 deg without changing the envelope magnitude.
10. A finite tube-root radius increases torque linearly through the first radial mass moment; inactive hub inertia reduces angular acceleration.

## 21. Conclusion

The specified six-blade tubular rotor is a balanced K-Field windmill. Each tube converts axial spin into a steady lateral force. Sixfold radial symmetry cancels the translational result while preserving the first radial moment of those forces as a continuous hub torque. The device therefore transforms the signed tube-force parity into controlled rotation about a mechanically free axis.

The orientation problem has an exact closed form. The global maximum is obtained by aligning the hub axis with G\_phase V\_E, not with the raw Earth-CMB vector. That configuration delivers 0.50833668 mN m for the stated geometry and spin speed, has no transverse torque, and is the cleanest actuator orientation. Requiring blades to cross the Earth-CMB line imposes a different geometric constraint and reduces useful torque to 0.26666996 mN m while adding a transverse reaction moment.

The one-degree sweep confirms the analytical result across the complete unit sphere. The torque field contains two antipodal maxima, a great-circle null, and opposite signed hemispheres. These features provide a direct experimental signature and a complete orientation prescription for construction, alignment, and testing.

## Appendix A. Complete numerical input ledger

| Parameter                                    | Value                 | Unit / meaning      |
|----------------------------------------------|-----------------------|---------------------|
| kappa_F recommended                          | 3.074050899702035e-11 | calibrated coupling |
| kappa_F lower                                | 3.042739944754650e-11 | calibration bound   |
| kappa_F upper                                | 3.105361854649420e-11 | calibration bound   |
| G_phase V_E                                  | 9.308252172704348e+05 | m/s                 |
| c_hat dot g                                  | 7.924608689771519e+05 | m/s                 |
| projection of g into CMB-perpendicular plane | 4.883045732394171e+05 | m/s                 |
| angle(c_hat,g_hat)                           | 31.64086127113854     | deg                 |
| blade count                                  | 6                     | integer             |
| tube OD                                      | 0.01                  | m                   |
| tube ID                                      | 0.008                 | m                   |
| tube length                                  | 0.1                   | m                   |
| density                                      | 8000                  | kg/m^3              |
| spin speed                                   | 50000                 | rpm                 |
| spin angular speed                           | 5235.987755982988     | rad/s               |

## Appendix B. Complete Cell3 basis and face-plane ledger

| Basis vector                      | ICRS x                | ICRS y               | ICRS z               | Length             |                    |
|-----------------------------------|-----------------------|----------------------|----------------------|--------------------|--------------------|
| a                                 | 0.0203814385169464    | -0.00341641449641055 | -0.00233296591672896 | 0.0207970587824329 |                    |
| b                                 | 0.00219874596832336   | 0.0129021702586925   | -0.0340665015360557  | 0.0364942051307272 |                    |
| c                                 | -0.000630379485347372 | -0.0242577220171279  | -0.0429243545710938  | 0.0493085658998594 |                    |
| Face family                       | Unit normal x         | Unit normal y        | Unit normal z        | d_plus             | d_minus            |
| ab faces (constant c coefficient) | 0.19409220080777      | 0.91317716882123     | 0.3583792347908      | -0.018828573500193 | 0.018828573500193  |
| ac faces (constant b coefficient) | 0.089052594101605     | 0.86657676929103     | -0.49103394832628    | 0.014052166903391  | -0.014052166903391 |
| bc faces (constant a coefficient) | -0.99596519030896     | 0.083602162507377    | -0.0326192905644     | -0.010254361619054 | 0.010254361619054  |

## Appendix C. All 27 centered Cell3 address nodes

| i  | j  | k  | ICRS x (native)       | ICRS y (native)      | ICRS z (native)      | i+j+k |
|----|----|----|-----------------------|----------------------|----------------------|-------|
| -1 | -1 | -1 | -0.0109749024999612   | 0.00738598312742294  | 0.0396619110119392   | -3    |
| -1 | -1 | 0  | -0.0112900922426349   | -0.004742877881141   | 0.0181997337263923   | -2    |
| -1 | -1 | 1  | -0.0116052819853086   | -0.0168717388897049  | -0.00326244355915457 | -1    |
| -1 | 0  | -1 | -0.00987552951579954  | 0.0138370682567692   | 0.0226286602439114   | -2    |
| -1 | 0  | 0  | -0.0101907192584732   | 0.00170820724820527  | 0.00116648295836448  | -1    |
| -1 | 0  | 1  | -0.0105059090011469   | -0.0104206537603587  | -0.0202956943271824  | 0     |
| -1 | 1  | -1 | -0.00877615653163786  | 0.0202881533861155   | 0.00559540947588353  | -1    |
| -1 | 1  | 0  | -0.00909134627431155  | 0.00815929237755154  | -0.0158667678096634  | 0     |
| -1 | 1  | 1  | -0.00940653601698523  | -0.00396956863101239 | -0.0373289450952103  | 1     |
| 0  | -1 | -1 | -0.000784183241487992 | 0.00567777587921766  | 0.0384954280535748   | -2    |
| 0  | -1 | 0  | -0.00109937298416168  | -0.00645108512934627 | 0.0170332507680279   | -1    |
| 0  | -1 | 1  | -0.00141456272683536  | -0.0185799461379102  | -0.00442892651751905 | 0     |
| 0  | 0  | -1 | 0.000315189742673686  | 0.0121288610085639   | 0.0214621772855469   | -1    |
| 0  | 0  | 0  | 0                     | 0                    | 0                    | 0     |
| 0  | 0  | 1  | -0.000315189742673686 | -0.0121288610085639  | -0.0214621772855469  | 1     |
| 0  | 1  | -1 | 0.00141456272683536   | 0.0185799461379102   | 0.00442892651751905  | 0     |
| 0  | 1  | 0  | 0.00109937298416168   | 0.00645108512934627  | -0.0170332507680279  | 1     |
| 0  | 1  | 1  | 0.000784183241487992  | -0.00567777587921766 | -0.0384954280535748  | 2     |
| 1  | -1 | -1 | 0.00940653601698523   | 0.00396956863101239  | 0.0373289450952103   | -1    |
| 1  | -1 | 0  | 0.00909134627431155   | -0.00815929237755154 | 0.0158667678096634   | 0     |
| 1  | -1 | 1  | 0.00877615653163786   | -0.0202881533861155  | -0.00559540947588353 | 1     |
| 1  | 0  | -1 | 0.0105059090011469    | 0.0104206537603587   | 0.0202956943271824   | 0     |
| 1  | 0  | 0  | 0.0101907192584732    | -0.00170820724820527 | -0.00116648295836448 | 1     |
| 1  | 0  | 1  | 0.00987552951579954   | -0.0138370682567692  | -0.0226286602439114  | 2     |
| 1  | 1  | -1 | 0.0116052819853086    | 0.0168717388897049   | 0.00326244355915457  | 1     |
| 1  | 1  | 0  | 0.0112900922426349    | 0.004742877881141    | -0.0181997337263923  | 2     |
| 1  | 1  | 1  | 0.0109749024999612    | -0.00738598312742294 | -0.0396619110119392  | 3     |

## Appendix D. Machine reconstruction equations

$$A_{\text{steel}} = \pi (R_o^2 - R_i^2)$$

$$m_{\text{blade}} = \rho A_{\text{steel}} L ; M = N m_{\text{blade}}$$

$$\mu_1 = \sum \text{integral } s \, dm = M L/2 \text{ for tubes beginning at the hub axis}$$

$$\omega_s = \text{RPM} (2 \pi/60)$$

$$g = G_{\text{phase}} V_E$$

$$F_{\text{tube}} = (\kappa_F m \omega_s/2) \hat{n} \text{ cross } g$$

$$\tau(\hat{h}) = -(\kappa_F \omega_s \mu_{1/4}) [g + (\hat{h} \text{ dot } g)\hat{h}]$$

$$\tau_h = -(\kappa_F \omega_s \mu_{1/2})(\hat{h} \text{ dot } g)$$

$$\tau_{\text{perp}} = -(\kappa_F \omega_s \mu_{1/4})[g - (\hat{h} \text{ dot } g)\hat{h}]$$

$$\tau_{\text{max}} = (\kappa_F \omega_s \mu_{1/2})|g|$$

$$|\tau_h| = \tau_{\text{max}} |\cos \gamma|$$

$$|\tau_{\text{perp}}| = (\tau_{\text{max}}/2)|\sin \gamma|$$

$$|\tau| = (\tau_{\text{max}}/2)\sqrt{1+3 \cos^2 \gamma}$$

$$h_{\text{peak}} = \pm g/|g|$$

$$h_{\text{cross}} = \pm [g - (\hat{c} \text{ dot } g)\hat{c}]/|g - (\hat{c} \text{ dot } g)\hat{c}|$$

$$I_{h,\text{blade}} = m [L^2/3 + (R_o^2 + R_i^2)/4]$$

$$\alpha_h = \tau_h/I_{\text{complete}}$$

$$P_h = \tau_h \Omega_h$$

## Appendix E. Orientation and torque result ledger

| Result                               | Value                                              |
|--------------------------------------|----------------------------------------------------|
| Global peak axis +                   | RA 189.285391176263 deg, Dec 16.626277960030 deg   |
| Global peak axis -                   | RA 9.285391176263 deg, Dec -16.626277960030 deg    |
| Global peak useful torque            | 5.083366763745302e-04 N m                          |
| Global peak interval                 | 5.031589785122050e-04 to 5.135143742368556e-04 N m |
| CMB vector                           | RA 167.943299879828 deg, Dec -6.942599922130 deg   |
| CMB separation from peak             | 31.640861271139 deg                                |
| CMB-crossing axis +                  | RA 250.207297826999 deg, Dec 47.867366474733 deg   |
| CMB-crossing axis -                  | RA 70.207297826999 deg, Dec -47.867366474733 deg   |
| CMB-crossing useful torque           | 2.666699603894507e-04 N m                          |
| CMB-crossing transverse torque       | 2.163869848058061e-04 N m                          |
| CMB-crossing total torque            | 3.434183963730822e-04 N m                          |
| Hub axis = CMB useful torque         | 4.327739696116122e-04 N m                          |
| Useful-torque null transverse torque | 2.541683381872651e-04 N m                          |
| Nearest 1-deg maximum                | RA 9 deg, Dec -17 deg, 5.083200843655000e-04 N m   |
| Nearest 1-deg null                   | RA 324 deg, Dec 67 deg, 1.696797162366573e-08 N m  |

## Appendix F. Source and file integrity ledger

| File                                                    | Role                                                       | SHA-256                                                          |
|---------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------|
| K-Field_tubular_rotor_force_analysis_1784897756.json    | Cell3 geometry, G_phase, Earth-CMB and calibrated coupling | 898e9433dc4930ef364392811bd324a629c1a8b7075d239b180aa05cc977e665 |
| K-Field_six_blade_windmill_torque_sphere_sweep_1deg.csv | 65,160-point one-degree orientation sweep                  | f3f097e6c33bd3d9bee2c784d91f5ebdc296bb43b8c7d5724ab3c714727b3d15 |

Machine-readable companion filename: K-Field\_tubular\_rotor\_earth\_CMB\_windmill\_1784901904.json

The companion JSON contains the complete input ledger, all derived values, all 27 Cell3 nodes, reference orientations, figure metadata, and the full one-degree torque sweep as columnar arrays.

**P. Dwayne Esterline, Principal Investigator**  
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